Scale Factors: Length, Area and Volume

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KEY WORDS scale factor *(tauwehe tauine)* · enlargement *(whakarahi)* · area *(horahanga)* · volume *(rōrahi)*

A Enlarging lengths ●○○

Remember: New length = old length × scale factor. A scale factor between 0 and 1 makes the shape smaller.

1. Find the new length.

a) 6 cm, scale factor 3 _____

b) 9 mm, scale factor 5 _____

c) 14 cm, scale factor 0.5 _____

d) 8 m, scale factor 1.5 _____

e) 20 cm, scale factor 2.5 _____

f) 36 mm, scale factor 0.25 _____

g) 4.5 m, scale factor 4 _____

h) 7.2 cm, scale factor 10 _____

2. Find the scale factor.

a) 4 cm becomes 12 cm _____

b) 5 mm becomes 35 mm _____

c) 10 cm becomes 25 cm _____

d) 16 m becomes 4 m _____

B Area scale factor ●●○

Remember: When lengths are multiplied by k, areas are multiplied by k^2.

3. Every length of a shape is multiplied by the scale factor. What is its area multiplied by?

a) scale factor 2

b) scale factor 3

c) scale factor 5

d) scale factor 10

e) scale factor 1.5

f) scale factor 0.5

4. Find the area of the enlarged shape.

a) Area 10 cm², scale factor 3

b) Area 7 m², scale factor 2

c) Area 4.5 cm², scale factor 10

d) Area 12 mm², scale factor 5

e) Area 80 cm², scale factor 0.5

f) Area 6 m², scale factor 1.5

C Volume scale factor ●●●

Remember: When lengths are multiplied by k, volumes are multiplied by k^3.

5. Every length of a solid is multiplied by the scale factor. What is its volume multiplied by?

a) scale factor 2

b) scale factor 3

c) scale factor 5

d) scale factor 10

e) scale factor 1.5

f) scale factor 0.5

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6. Find the volume of the enlarged solid.

a) Volume 20 cm³, scale factor 2

b) Volume 5 m³, scale factor 3

c) Volume 1.2 cm³, scale factor 10

d) Volume 6 mm³, scale factor 4

e) Volume 400 cm³, scale factor 0.5

f) Volume 16 m³, scale factor 1.5

D Finding the scale factor ●●●

⚠ **Watch out:** The scale factor for lengths is the square root of the area scale factor, or the cube root of the volume scale factor.

7. Find the scale factor k for the lengths.

a) Two similar shapes have areas of 5 cm² and 45 cm².

b) Two similar shapes have areas of 8 m² and 200 m².

c) Two similar solids have volumes of 4 cm³ and 108 cm³.

d) Two similar solids have volumes of 10 m³ and 80 m³.

e) Two similar shapes have areas of 12 cm² and 27 cm².

f) Two similar solids have volumes of 1,000 mm³ and 125 mm³.

E Problems ●●●

8. A model car is made using a scale of 1 : 20. The real car is 4.2 m long. How long is the model, in centimetres?

9. A photo is 10 cm wide and 15 cm high. It is enlarged with a scale factor of 2.5.

a) Find the width and height of the enlargement.

b) Find the area of the enlargement.

10. A small pizza has a diameter of 20 cm. A large pizza has a diameter of 40 cm. How many times more pizza is in the large one?

11. A stone statue is 2 m tall and has a mass of 600 kg. A model of it, made of the same stone, is 20 cm tall. Find the mass of the model.

★ Extension

12. Two similar cylinders have surface areas of 50 cm² and 450 cm². The smaller cylinder has a volume of 30 cm³. Find the volume of the larger cylinder.

13. Every edge of a cube is made 50% longer. By what percentage does the volume of the cube increase?

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Scale Factors: Length, Area and Volume

Name _____ Date _____ Calculator allowed Version 2

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KEY WORDS scale factor *(tauwehe tauine)* · enlargement *(whakarahi)* · area *(horahanga)* · volume *(rōrahi)*

A Enlarging lengths ●○○

Remember: New length = old length × scale factor. A scale factor between 0 and 1 makes the shape smaller.

1. Find the new length.

a) 7 cm, scale factor 4 _____

b) 8 mm, scale factor 6 _____

c) 18 cm, scale factor 0.5 _____

d) 6 m, scale factor 1.5 _____

e) 12 cm, scale factor 2.5 _____

f) 48 mm, scale factor 0.25 _____

g) 3.5 m, scale factor 4 _____

h) 6.4 cm, scale factor 10 _____

2. Find the scale factor.

a) 3 cm becomes 15 cm _____

b) 6 mm becomes 24 mm _____

c) 10 cm becomes 35 cm _____

d) 20 m becomes 4 m _____

B Area scale factor ●●○

Remember: When lengths are multiplied by k, areas are multiplied by k^2.

3. Every length of a shape is multiplied by the scale factor. What is its area multiplied by?

a) scale factor 2

b) scale factor 4

c) scale factor 6

d) scale factor 10

e) scale factor 2.5

f) scale factor 0.5

4. Find the area of the enlarged shape.

a) Area 10 cm², scale factor 4

b) Area 9 m², scale factor 2

c) Area 2.5 cm², scale factor 10

d) Area 15 mm², scale factor 3

e) Area 60 cm², scale factor 0.5

f) Area 8 m², scale factor 1.5

C Volume scale factor ●●●

Remember: When lengths are multiplied by k, volumes are multiplied by k^3.

5. Every length of a solid is multiplied by the scale factor. What is its volume multiplied by?

a) scale factor 2

b) scale factor 4

c) scale factor 6

d) scale factor 10

e) scale factor 2.5

f) scale factor 0.5

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6. Find the volume of the enlarged solid.

a) Volume 30 cm³, scale factor 2

b) Volume 4 m³, scale factor 3

c) Volume 2.4 cm³, scale factor 10

d) Volume 3 mm³, scale factor 5

e) Volume 800 cm³, scale factor 0.5

f) Volume 8 m³, scale factor 1.5

D Finding the scale factor ●●●

⚠ **Watch out:** The scale factor for lengths is the square root of the area scale factor, or the cube root of the volume scale factor.

7. Find the scale factor k for the lengths.

a) Two similar shapes have areas of 6 cm² and 96 cm².

b) Two similar shapes have areas of 7 m² and 175 m².

c) Two similar solids have volumes of 5 cm³ and 135 cm³.

d) Two similar solids have volumes of 3 m³ and 192 m³.

e) Two similar shapes have areas of 20 cm² and 45 cm².

f) Two similar solids have volumes of 1,000 mm³ and 216 mm³.

E Problems ●●●

8. A model car is made using a scale of 1 : 25. The real car is 4.5 m long. How long is the model, in centimetres?

9. A photo is 12 cm wide and 18 cm high. It is enlarged with a scale factor of 1.5.

a) Find the width and height of the enlargement.

b) Find the area of the enlargement.

10. A small pizza has a diameter of 15 cm. A large pizza has a diameter of 45 cm. How many times more pizza is in the large one?

11. A stone statue is 3 m tall and has a mass of 1,350 kg. A model of it, made of the same stone, is 30 cm tall. Find the mass of the model.

★ Extension

12. Two similar cylinders have surface areas of 20 cm² and 320 cm². The smaller cylinder has a volume of 15 cm³. Find the volume of the larger cylinder.

13. Every edge of a cube is made 20% longer. By what percentage does the volume of the cube increase?

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